

Vendor Performance Analysis

Retail Inventory & Sales

Vendor efficiency and profitability for strategic purchasing and inventory decisions,
with machine-learning models for freight cost and invoice manual approval

Period	2024-01-01 to 2024-12-31
Purchase orders analysed	5,403 (in period)
Vendors	126 with purchases
Purchases	\$311.61M
Tools	SQLite, Python (pandas, scikit-learn, SciPy), Plotly, Streamlit

1. Executive summary

We analysed 5,403 purchase orders from 126 vendors (\$311.61M of purchases), scored every vendor on profitability and efficiency, and built two models: one that predicts the freight an invoice should carry, and one that flags invoices that should go to manual approval.

- **Purchases are concentrated.** The top 10 vendors account for 65.3% of spend, while the bottom 63 vendors supply only 0.7%.
- **The biggest suppliers earn below-median margins.** 34 'Volume driver' vendors carry 56% of spend: cost renegotiation here has the largest profit effect.
- **Inventory grew 18%** (\$46.91M to \$55.43M at cost). 23 underperforming vendors hold \$1.86M of it.
- **Freight is a flat 0.5% of invoice value**, and no ML model beats that rule. The model is valuable as a benchmark: it exposed 86 overcharged invoices (about \$32.9K excess) all dated 2024-01-04 to 2024-01-12.
- **The approval control has a gap.** Manual approval is triggered only by invoice size ($\geq$ \$250K). 206 anomalous invoices (\$2.58M) were never reviewed. The risk model catches 99% of invoices that need review at 97% precision on held-out data, versus 65% recall for the dollar rule alone.
- **Important limitation:** no sales data or PO line items were available, so units sold, revenue and profit are estimates (see section 3).

Recommended actions

- Add anomaly-based triggers (freight above 1.5x expected, unit cost far from the vendor norm) to the \$250K approval rule.
- Investigate and recover the early-January freight overcharges; monitor freight against the 0.5% expected rate.
- Focus renegotiation on the high-volume, below-median-margin vendors; review the underperformer vendors.
- Release cash from slow stock, starting with high-margin SKUs whose inventory is growing.
- Consolidate the long tail of small vendors while managing dependence on the top 10.

2. Business problem and data

A retailer buys from many vendors. The questions: which vendors create the most value, where is capital tied up in slow stock, are freight charges reasonable, and which invoices deserve extra scrutiny?

File	Content	Rows
export_output_vendor_invoice.csv	One row per PO invoice: vendor, dates, quantity, dollars, freight, approver	5,543
export_output_purchase_prices.csv	Product catalog: brand, vendor, retail and purchase price	12,261
export_output_begin_inventory.csv	On-hand units by store and brand, 2024-01-01	206,529
export_output_end_inventory.csv	On-hand units by store and brand, 2024-12-31	224,489

3. Method

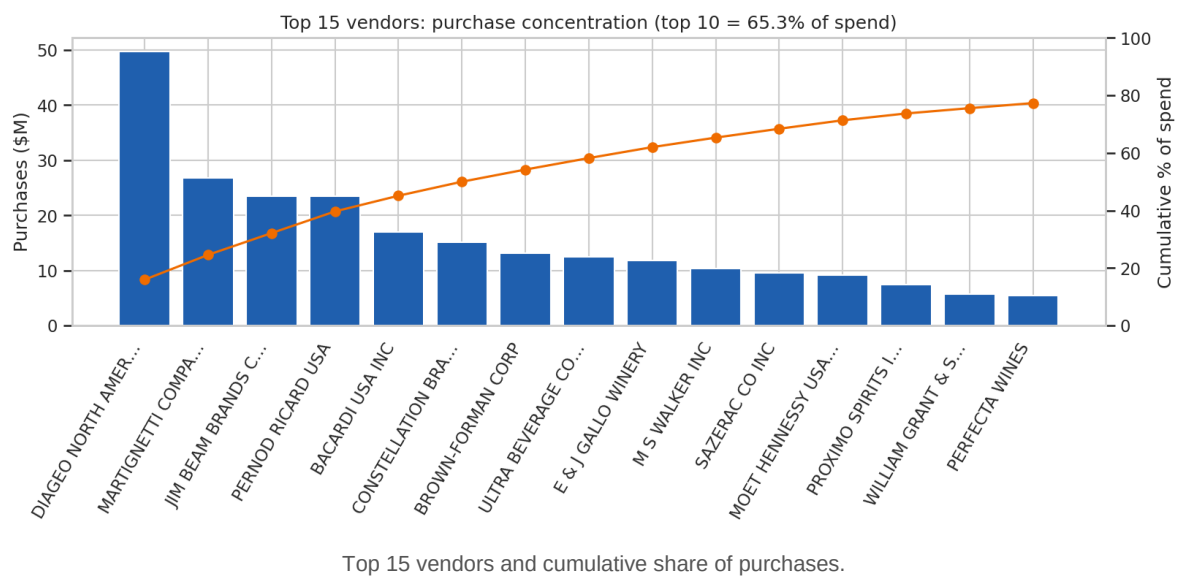
Pipeline. Raw CSVs are loaded into SQLite; SQL (CTEs, joins, window functions) builds the PO-level purchases table (*export_output_purchases.csv*), the SKU table and the vendor summary. Pandas adds KPIs, a composite vendor score and segments. Every build asserts that totals reconcile to the raw files.

Estimated sales (limitation). Without a sales file, units sold per vendor are estimated as begin inventory + units purchased - end inventory; revenue and gross profit use the inventory-weighted shelf price and catalog purchase price. Vendors with fewer than 3 POs or 100 units are marked unreliable and excluded from sales-based KPIs. Invoice date stands in for receiving date.

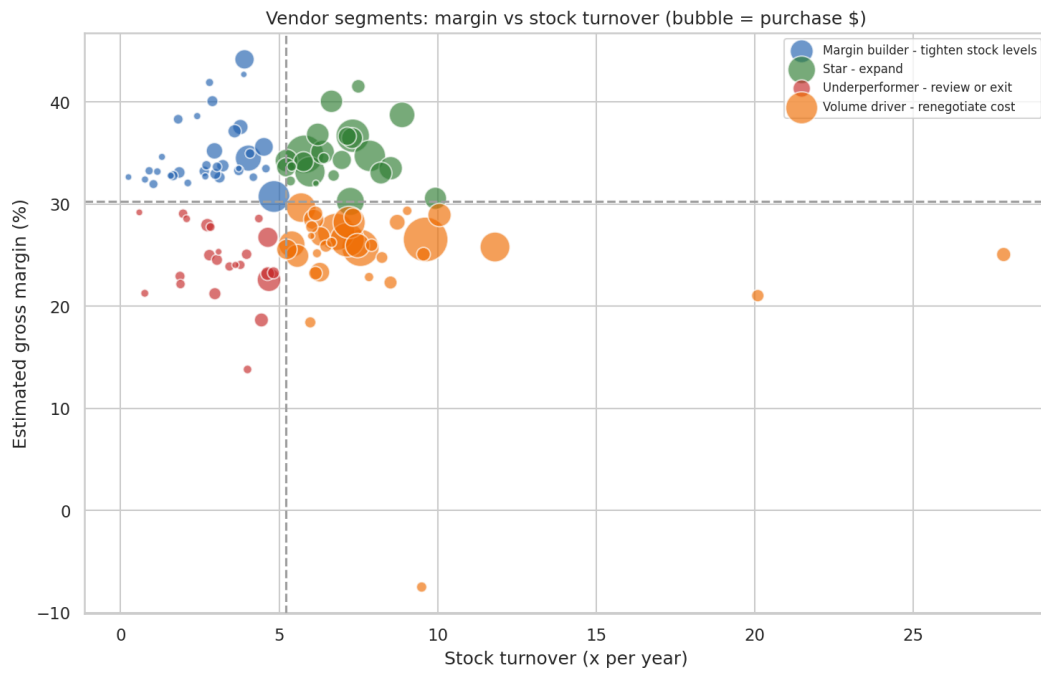
Vendor score and segments. A weighted percentile-rank score (margin 35%, sell-through 20%, turnover 20%, lead time 7%, lead-time consistency 6%, freight % 7%, payment terms 5%). Vendors are placed in a margin x turnover quadrant: Star, Margin builder, Volume driver, Underperformer.

Data quality. The raw data is structurally clean. Minor issues handled: two vendors renamed mid-year, one invoiced vendor missing from the catalog, six catalog vendors never invoiced, two SKUs with zero pricing, and 140 POs invoiced after year-end (kept, excluded from vendor KPIs).

4. Vendor performance findings



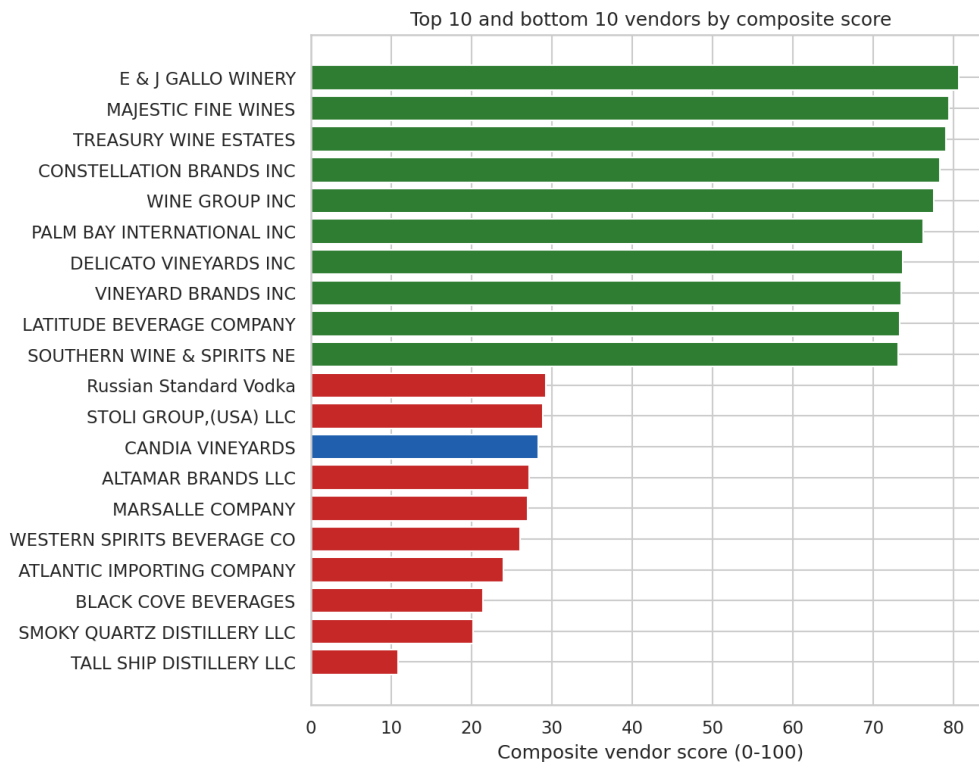
The top 10 vendors (Diageo, Martignetti, Jim Beam, Pernod Ricard, Bacardi, ...) make up 65.3% of purchases (HHI 604, an unconcentrated market overall). That gives negotiating leverage but also dependence risk.



Vendor segments by margin and stock turnover (bubble size = purchases).

Segment	Vendors	Share of spend	Action
Star	24	34.3%	Expand
Margin builder	34	7.6%	Tighten stock levels
Volume driver	34	55.8%	Renegotiate cost
Underperformer	23	2.4%	Review or exit
Insufficient data	17	0.01%	Not scored

Highest composite scores: E & J Gallo Winery, Majestic Fine Wines, Treasury Wine Estates, Constellation Brands Inc, Wine Group Inc. Lowest: Western Spirits Beverage Co, Atlantic Importing Company, Black Cove Beverages, Smoky Quartz Distillery Llc, Tall Ship Distillery Llc.



Top 10 and bottom 10 vendors by composite score.

Research questions

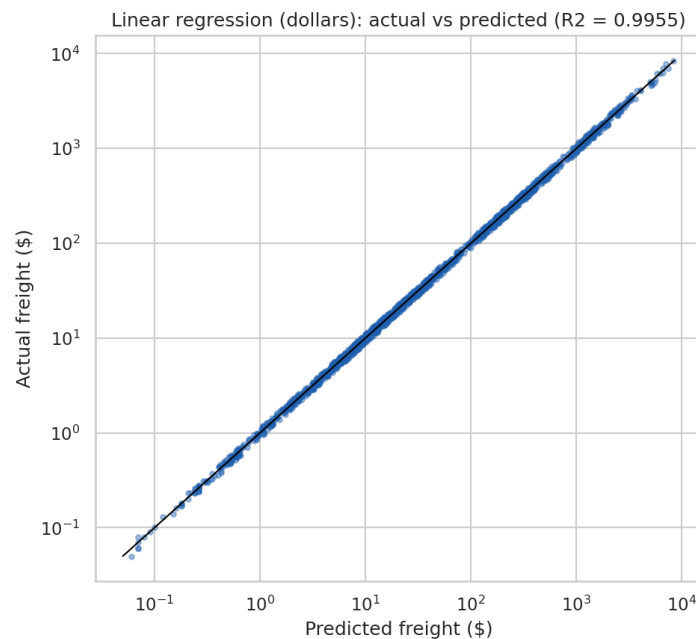
- **Brands to promote or reprice:** 377 SKUs combine a top-quartile margin (average 41%), growing inventory and a large stock value (\$5.64M at cost). Brand-level sales are unavailable, so growing stock is the proxy for slow movement.
- **Capital tied up:** inventory at cost grew 18.2%. The 23 underperformers hold \$1.86M on 2.4% of spend.
- **Does bulk buying cut unit cost?** No evidence. Within a vendor, median unit cost of the largest orders is 0.995x the vendor average vs 0.997x for the smallest. Freight is also about 0.51% at every order size.
- **Do high-spend vendors earn different margins?** No (Welch t-test: 30.9% vs 30.1%, difference 0.87 pp, p = 0.55, 95% CI [-2.0, 3.8] pp). Vendor size does not predict profitability.

5. Model 1: freight cost prediction

Goal. Estimate the freight an invoice should carry from information known when it arrives (order value, quantity, unit cost, lead time, payment terms, calendar, the vendor's typical rate). **Design.** Temporal hold-out: trained on invoices before 2024-10-01 (3,893 clean rows), tested on 1,564 later invoices. Candidates: the 0.5% rule baseline, linear regression, ridge, random forest and gradient boosting; the simplest ML model within 2% of the best cross-validated MAE is selected.

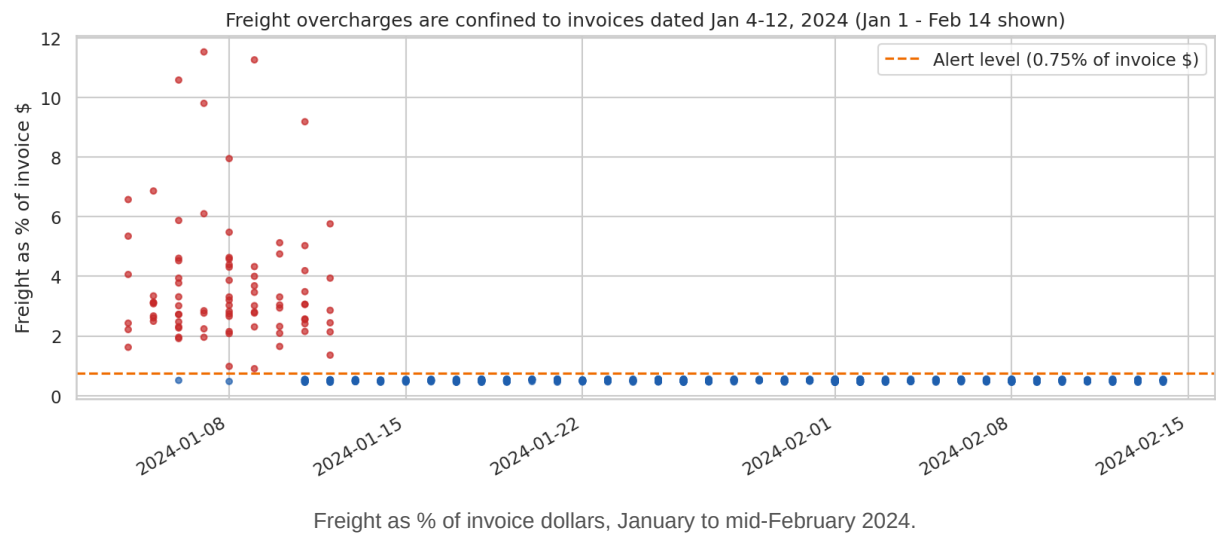
Model	CV MAE	Test MAE	Test R2	Median % err
Baseline (median rate x dollars)	\$15.79	\$17.14	0.9955	5.6%
Linear regression (dollars)	\$15.84	\$17.15	0.9955	5.6%
Ridge (dollars, quantity)	\$15.89	\$17.52	0.9955	5.1%
Random forest (rate)	\$15.79	\$17.84	0.9950	5.3%
Gradient boosting (rate)	\$16.41	\$16.88	0.9962	5.3%

Result. Every model reaches R^2 of about 0.9955. None beats the one-line rule 'freight = 0.50% of invoice dollars' (test MAE \$17.15 vs \$17.14), because freight is proportional to order value and the other features add nothing. The selected model is therefore the simple, interpretable one: *Freight = 0.00500 x Dollars (no intercept)*. This is a useful negative result: a complex model would add cost and opacity for no accuracy.



Hold-out actual vs predicted freight (log scales).

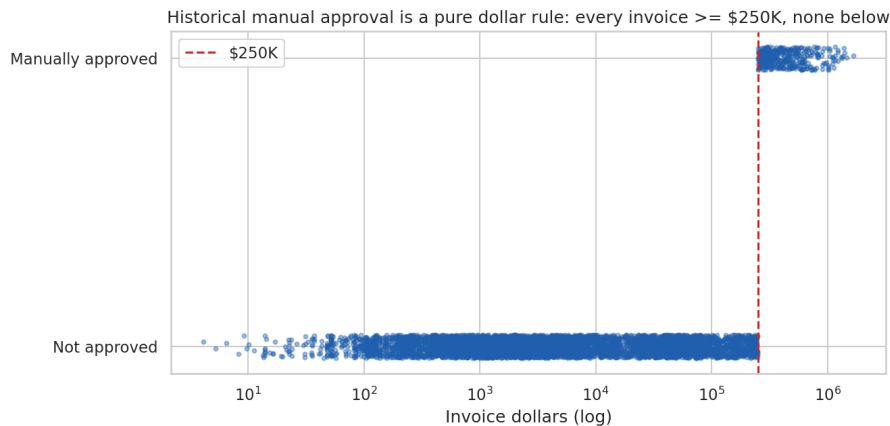
Freight overcharges



The expected-freight benchmark exposes 86 invoices with freight above 1.5x the standard rate, a median of 6.1x the expected amount and about **\$32.9K** of excess freight. All fall on invoices dated 2024-01-04 to 2024-01-12, and 80% of invoices in that window are affected: a systematic billing event (for example a surcharge or a wrong rate table) rather than random noise. Because it occurs in one window, the freight model cannot be validated on overcharges in later months.

6. Model 2: invoice manual-approval prediction

Finding that shaped the design. Historical manual approval is a pure dollar rule: all 374 invoices of about \$250.7K or more were approved (by one approver) and none below \$250K were. A model trained on that label alone would trivially score 100% by learning one threshold, so the target was widened to include anomaly flags.

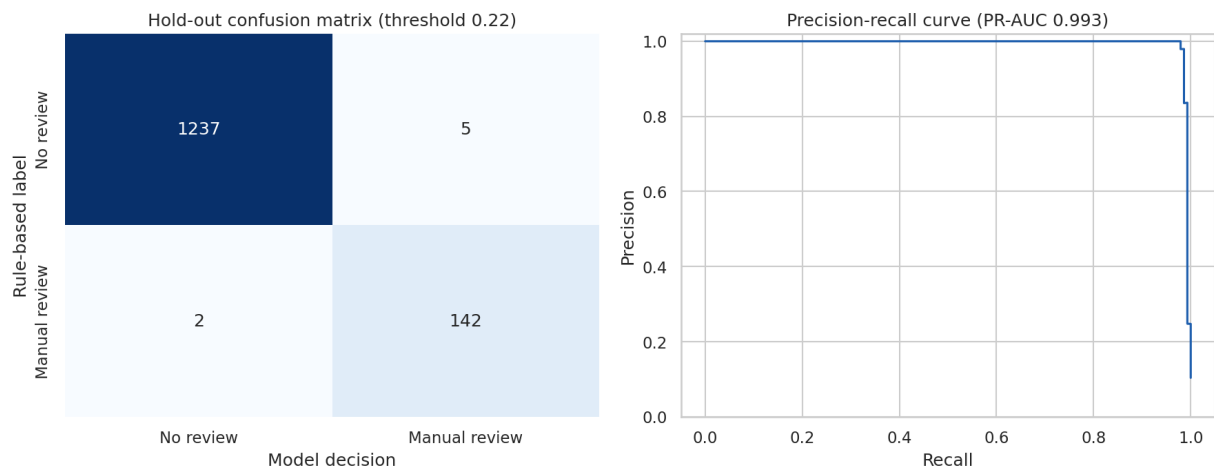


Approval versus invoice size: a clean threshold at \$250K.

Target (*ManualApprovalRequired*): any of the following.

Flag	Rule	Invoices
High value	Invoice $\geq$ \$250K (observed policy)	374
Freight overcharge	Freight above 1.5x the standard 0.5% rate	86
Unit-cost outlier	Unit cost $\geq$ 2x or $\leq$ 0.5x the vendor median	112
Lead-time outlier	$ \text{robust } z > 4$ versus the vendor norm	8
Payment-terms outlier	$ \text{robust } z > 4$ versus the vendor norm	2
Repeat invoice	Same vendor, dollars, quantity within 14 days	2
Needs manual approval	Any flag	580 of 5,543

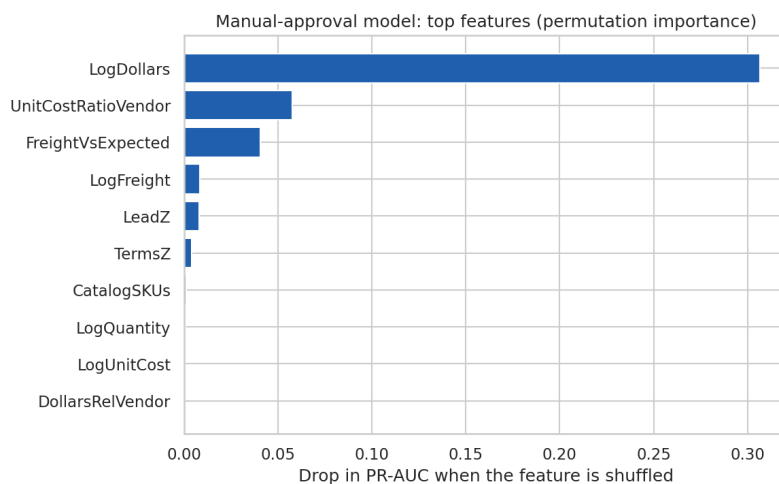
Model. The classifier sees only invoice values and how they compare with the vendor's norms (computed on training data only) plus the freight model's expected freight; never the flags or the approval column. Logistic regression, random forest and gradient boosting were compared by 5-fold cross-validated PR-AUC; **Random forest** was selected and its threshold set by maximising F2 (a missed risky invoice costs more than an unneeded review).



Hold-out confusion matrix and precision-recall curve.

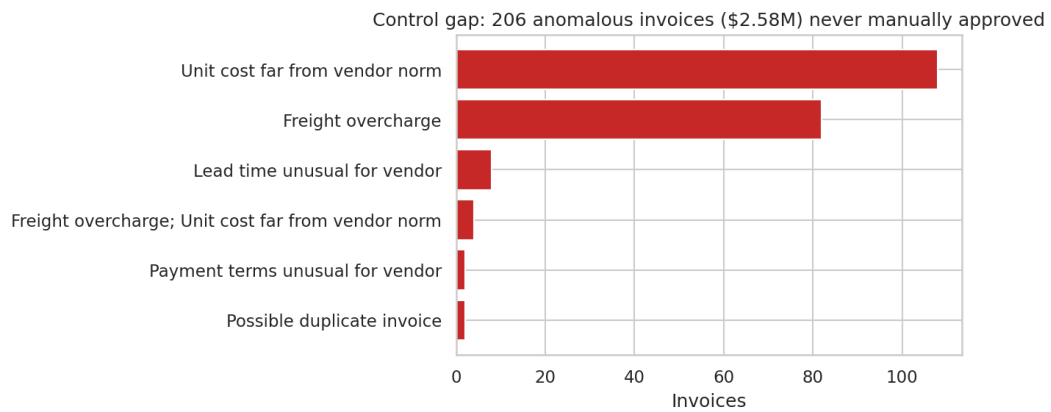
Evaluation	Precision	Recall	Note
Hold-out (25% of invoices)	0.966	0.986	PR-AUC 0.993
Invoices below \$250K only	0.906	0.960	Anomaly detection beyond the dollar rule
Time-based split (train before Oct)	0.972	0.983	Robustness over time
\$250K rule alone (current control)	1.000	0.653	Misses the anomalies
Isolation Forest (unsupervised)	0.465	0.465	ROC-AUC 0.90; much weaker

How to read this honestly. The labels are rule-derived (weak supervision), not confirmed fraud. The scores show that the review policy is learnable and can be applied to new invoices, with a probability and reasons attached; they do not prove independent fraud detection. The very rare flags (lead time, payment terms, repeat invoices) have only a handful of examples and are not reliably learned; the rules still catch them.



Top features by permutation importance.

Control gap



Anomalous invoices below \$250K that never received manual approval.

206 invoices below \$250K carried at least one anomaly flag but were never manually approved (\$2.58M of purchases). Adding the anomaly triggers to the approval policy would send roughly 206 more invoices a year to review (3.7% of all invoices), a manageable workload.

7. Recommendations

- **Close the approval control gap** by adding freight and unit-cost anomaly triggers to the \$250K rule; use the risk model to rank the review queue and show the reasons.
- **Recover January freight overcharges** (\$32.9K across 86 invoices) and monitor freight against the 0.5% expected rate.
- **Renegotiate with the Volume drivers** (56% of spend at below-median margin): a small cost reduction has the largest profit effect.
- **Release cash from slow stock**: start with the 377 high-margin SKUs with growing inventory and review the 23 underperformer vendors.
- **Consolidate the vendor tail** (63 vendors supply 0.7% of spend) while managing the top-10 dependence.
- **Do not budget on volume discounts or freight savings from consolidation**: neither appears in the data.

8. Limitations

- No sales or PO line-item data: units sold, revenue and profit are inventory-balance estimates; vendors with thin data are excluded from sales KPIs.
- Invoice date stands in for receiving date; lead time is PO-to-invoice, so on-time delivery cannot be measured.
- Approval labels are rule-derived; the model operationalises the policy and is not independent fraud detection.
- Freight overcharges occur in a single nine-day window, so they cannot be validated across time.
- Promotion candidates use growing stock as a proxy for slow sales; confirm with sales data before acting.

9. Deliverables

GitHub-ready project: ingestion and analysis scripts, two executed notebooks (EDA and analysis with models), trained models, a Streamlit dashboard with a freight estimator and an invoice checker, tests, and this report. See README.md for how to run everything.